

DECODELABS · INDUSTRIAL TRAINING KIT

Project 4: Data Visualization

Order Performance Dashboard — Data Storytelling Report

Built with Power BI | Dataset: 1,200 orders, Jan 2023 – Jun 2025

Batch 2026

Prepared by: R.Dunith Nadupa

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1. Objective

Project 4 asks for visual representations of the order dataset that communicate insights clearly — not just charts, but data storytelling. The specific requirements are to create appropriate charts (bar, line, etc.), choose the right visual for each type of data, and highlight the key insights so a decision-maker can act on them within seconds.

This report follows the three disciplines outlined in the brief:

- The Architect — choosing the chart type that strictly matches the business question.
- The Editor — maximizing the data-ink ratio and removing chartjunk, legends, and 3D distortion.
- The Storyteller — structuring every page around the Situation → Complication → Resolution (SCR) narrative, with an action title that states the conclusion.

2. Dataset Overview

The dataset contains 1,200 individual e-commerce orders placed between January 2023 and June 2025, across 1,189 unique customers.

Field	Description
OrderID / CustomerID	Unique identifiers for each order and customer
Date	Order date (Jan 2023 – Jun 2025)
Product	One of 7 categories: Chair, Printer, Laptop, Tablet, Monitor, Desk, Phone
Quantity, UnitPrice, TotalPrice	Order line economics — $TotalPrice = Quantity \times UnitPrice$
PaymentMethod	Cash, Online, Credit Card, Debit Card, Gift Card
OrderStatus	Delivered, Shipped, Pending, Cancelled, Returned
CouponCode / ReferralSource	Marketing attribution fields

3. The Dashboard

The dashboard is built as four report pages, each dedicated to a single message — per the brief's rule that "one slide, one message" prevents crowded pages from diluting insights. Pages 1–3 follow the SCR framework; page 4 is supporting channel detail.

Page 1 — Situation

ACTION TITLE

"Revenue reached \$1.26M across 1,200 orders, but quarterly growth has cooled since its 2023 peak"

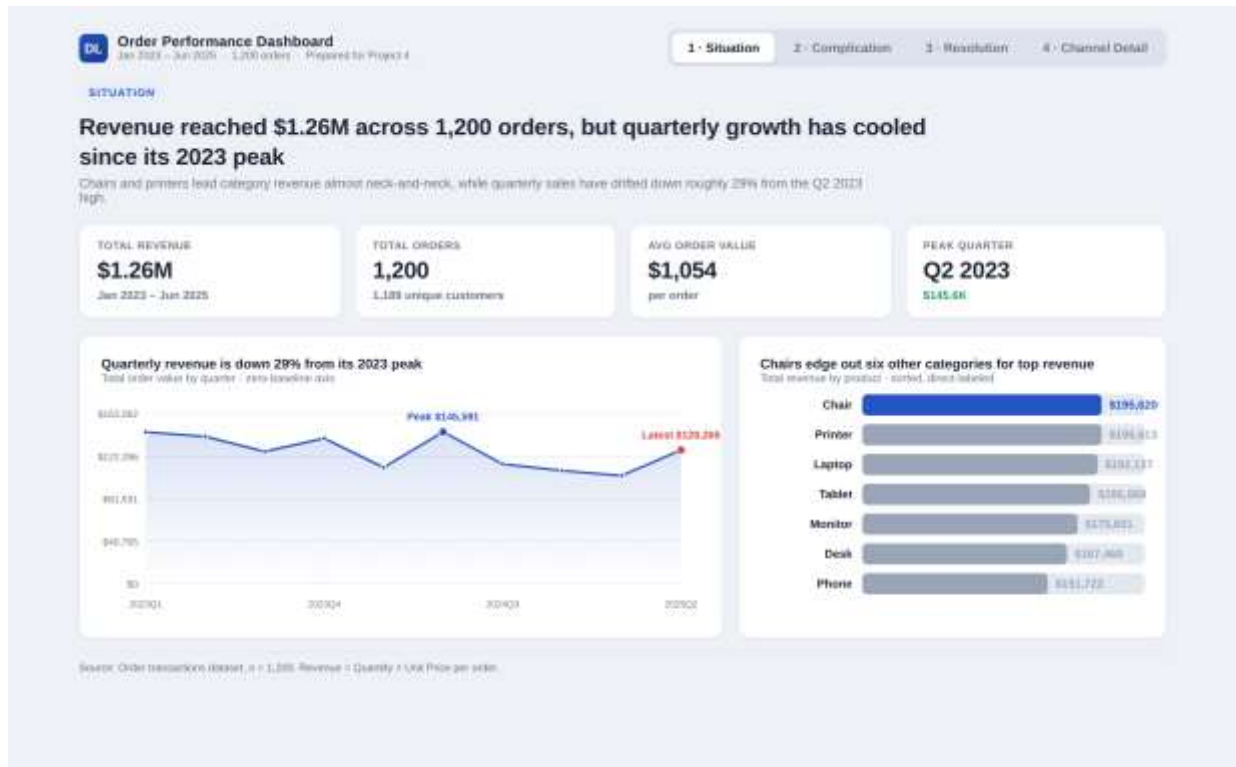


Fig. 1 — KPI cards (top-left, per the 5-second rule) plus a zero-baseline revenue trend and a direct-labeled, legend-free product ranking.

Chart choices:

- Line chart for the quarterly trend — the correct choice for showing change over a continuous time axis.
- Horizontal bar chart for revenue by product — best for comparing values across named categories, sorted descending, with data labels replacing the legend.

Insight: Quarterly revenue peaked at \$145.6K in Q2 2023 and has declined to \$128.3K by Q2 2025 — a 29% drop from peak. Chair and Printer are effectively tied for the top revenue category, with Phone lowest.

Page 2 — Complication

ACTION TITLE

"41% of all orders are Cancelled or Returned — \$520K in order value never converts to kept revenue"

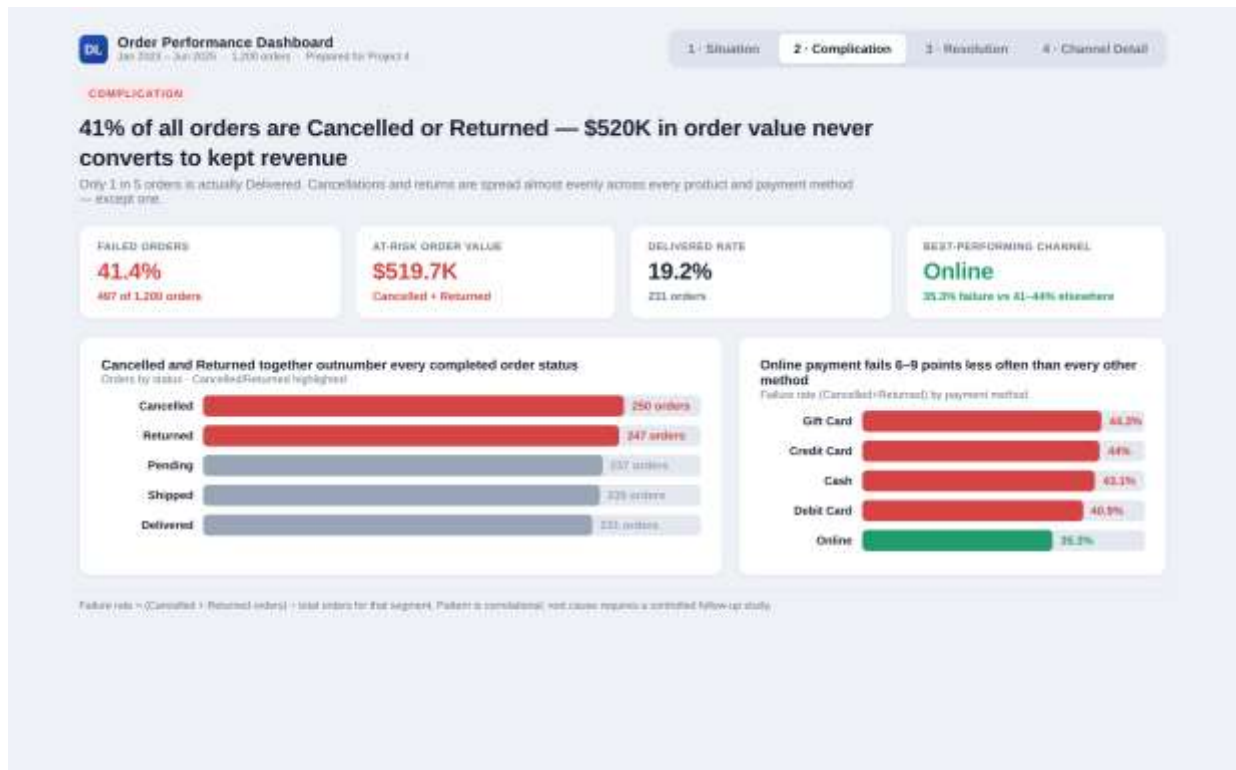


Fig. 2 — Order status ranked by volume (Cancelled/Returned spotlighted in red) and failure rate by payment method (Online spotlighted in green).

Chart choices:

- Horizontal bar chart for order status — a pie chart was deliberately avoided here, since the brief rules out pie charts above 3 slices and OrderStatus has 5.
- Color used only as a spotlight: red marks the two failure states, grey holds the rest as context — never decoration.

Insight: Only 19.2% of orders reach Delivered. Cancelled (20.8%) and Returned (20.6%) combined outweigh every completed status. Failure is broadly consistent across payment methods (41–44%) except Online, which fails at just 35.3%.

Page 3 — Resolution

ACTION TITLE

"Matching every payment method to Online's 35.3% failure rate could recover ~\$73K in order value"

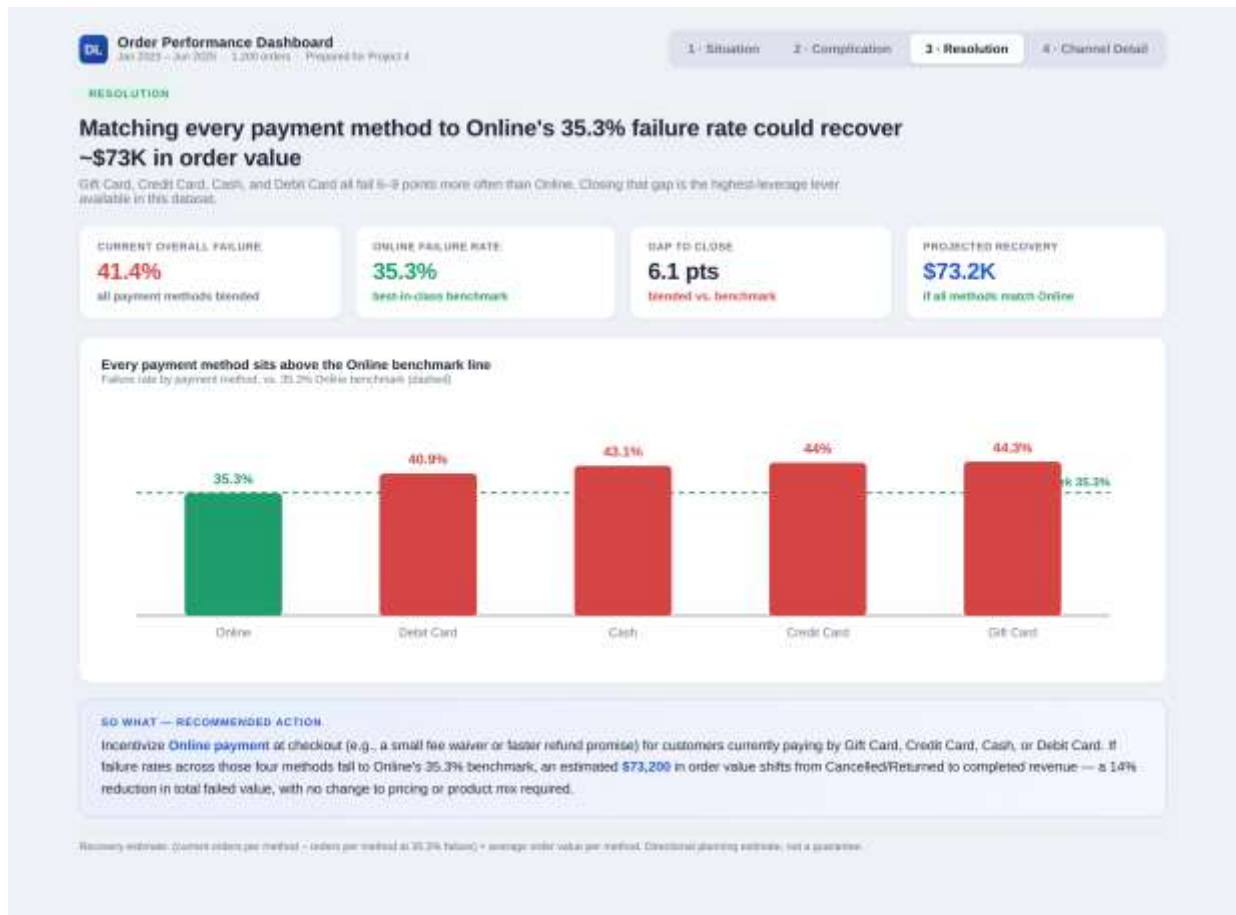


Fig. 3 — Failure rate by payment method against the Online benchmark line, with the data-driven recommendation stated directly on the page.

Chart choices:

- A benchmark (constant) line makes the gap between each payment method and the best performer immediately visible, without requiring the reader to do mental subtraction.
- The recommendation is stated as text on the page itself — the brief requires every page to answer "so what should we do next?", not just present the number.

Insight: If Gift Card, Credit Card, Cash, and Debit Card orders failed at Online's 35.3% rate instead of their current 41–44%, an estimated \$73,200 in order value would convert instead of being lost — a 14% reduction in total failed value.

Page 4 — Channel Detail (Supporting)

ACTION TITLE

"Instagram drives the most total revenue, but Facebook customers spend the most per order"



Fig. 4 — Revenue and average order value by acquisition channel, kept visually quieter than pages 1–3 since it is supporting detail, not the primary narrative.

Insight: All five channels perform within a tight band. Instagram leads on total revenue (\$275.3K) largely on volume, while Facebook has the highest average order value (\$1,098) despite lower volume — two different levers for two different marketing goals.

4. Requirements Checklist

Mapped directly against the Key Requirements and Key Skills stated in the Project 4 brief:

✓ **Create charts (bar, line, pie, etc.)**

Line chart (trend), horizontal bar charts (comparison, ranking, benchmark) used across all 4 pages.

✓ **Choose appropriate visuals for the data**

Trend → line; category comparison → bar; no pie chart used anywhere, since every categorical field here has 5+ categories.

✓ **Highlight key insights**

One spotlight accent color per page (blue/red/green) isolates the single data point that proves each page's action title.

✓ **Axis integrity**

Every bar and line chart starts its value axis at zero — no truncated or distorted axes.

✓ **Data-ink ratio / no chartjunk**

Legends replaced with direct labels; no 3D effects, heavy borders, or decorative gridlines.

✓ **Data storytelling (SCR framework)**

Pages 1–3 move through Situation → Complication → Resolution; page 4 is clearly marked as supporting detail.

5. Executive Summary

Order Performance Review — January 2023 to June 2025

Situation: The business generated \$1.26M in revenue across 1,200 orders (1,189 unique customers, \$1,054 average order value), but quarterly revenue has declined roughly 29% from its Q2 2023 peak of \$145.6K to \$128.3K in Q2 2025.

Complication: 41.4% of all orders — 497 of 1,200 — end in Cancelled or Returned status, representing \$519,700 in order value that never converts to kept revenue. Only 19.2% of orders reach Delivered. This pattern holds across products (39–44% failure) and across four of five payment methods (41–44%), with one exception: Online payment fails at only 35.3%.

Resolution: Shifting order volume toward Online payment — and closing the 6.1-point gap to its benchmark failure rate — is projected to recover approximately \$73,200 in order value, a 14% reduction in total failed value, without any change to pricing or product assortment.

This dataset shows correlation, not proven causation, between payment method and order failure. The recommendation is a data-driven hypothesis worth testing (e.g., a checkout incentive A/B test), not a guaranteed outcome.

6. Tools & Method

Dashboard built in Power BI, using DAX measures for Total Revenue, Failed Order Rate, Online Failure Rate, and Projected Recovery Value, on top of the cleaned order dataset. Color theme limited to 5 hues (navy ink, muted grey, and one accent per narrative beat — blue for growth, red for risk, green for resolution), per the brief's spotlight-color rule.